RetroInv

The Retro Computer Collection Inventory



Owner's Manual

For the Apple IIe, Apple IIc and Apple IIGS

RetroInv

Owner's Manual · Version 2.5

The Retro Computer Collection Inventory

By 6502Retro

For the enhanced Apple IIe with an 80-column card, the Apple IIc, the Apple IIc Plus and the Apple IIGS, running ProDOS 8. A companion Applesoft BASIC version runs on any Apple II with 48K.

Notice

6502Retro reserves the right to make improvements to the program described in this manual at any time and without notice.

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The software is provided "as is." Should RetroInv prove to have a bug, you, the collector, assume the entire cost of re-typing your Disk II serial numbers. 6502Retro does, however, promise to take your bug report seriously and to fix it as fast as a 1 MHz 6502 allows.

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About This Manual

RetroInv keeps track of your vintage computer collection: every machine, card, disk and manual, with its maker, serial number, year, condition, shelf location and value. This manual shows you how to use it.

You don't need to be an expert. If you can start a program from ProDOS and press a few keys, you can use RetroInv. Chapters 1 and 2 get you started and show you around. Chapters 3 through 7 cover each part of the program in turn, and the appendixes are there when you need to look something up.

Conventions Used in This Manual

- **Keys** appear in small capitals, the way they're labeled on your keyboard: RETURN, ESC, TAB, DELETE, and the arrow keys ↑ ↓ ← →.
- **Open-Apple** (⌘) is the key with the outlined apple, to the left of the Space bar. "Press ⌘-S" means hold down Open-Apple and press S.
- CONTROL-**X** means hold down CONTROL and press X.
- **Menu commands** are written like this: choose Save from the File menu.
- **What you type** appears in plain capitals: type `INVENTORY`.
- **MouseText** is the set of special characters, such as the pointer ↖, the window borders and the scroll bar, that the enhanced Apple IIe, the IIc and the IIGS can show in text mode. RetroInv uses MouseText to draw its windows.



By the Way: Notes like this one hold tips and extra information. You can skip them and still use RetroInv, but they often save you time.
Important! Notes like this one tell you something you need to know to avoid trouble.
Warning: Notes like this one tell you how to avoid losing information.

CHAPTER 1

Getting Started

This chapter tells you what equipment RetroInv needs, how to make a backup copy of the RetroInv disk, and how to start the program.

What You Need

ITEM	REQUIRED?	NOTES
Enhanced Apple IIe with an 80-column card, Apple IIc, IIc Plus or IIGS	Yes	RetroInv needs MouseText and 80 columns
ProDOS 8	Yes	ProDOS 8 version 2.4.3 is on the RetroInv disk
A disk drive	Yes	A 5.25-inch drive, a 3.5-inch drive or a hard disk
AppleMouse II, or the IIc or IIGS mouse port	No	Everything works from the keyboard too
Mouse Calc 1.1	No	Only for Chapter 7



Important! An Apple II, II Plus or unenhanced Apple IIe can't show MouseText. On those machines RetroInv displays a message and returns you to ProDOS. Use the Applesoft version instead (see Chapter 6).

What's on the RetroInv Disk

The RetroInv disk is a ProDOS disk named `/RETROINV`. It holds these files:

FILE	TYPE	WHAT IT IS
`PRODOS`	SYS	ProDOS 8, version 2.4.3
`RETROINV.SYSTEM`	SYS	The RetroInv program
`BASIC.SYSTEM`	SYS	Applesoft BASIC for ProDOS, for the Applesoft version
`RETROINV`	BAS	The Applesoft BASIC version

FILE	TYPE	WHAT IT IS
SAMPLE	TXT	A sample inventory with two complete computer systems
SAMPLE.INV	$F8	The same sample as a Mouse Calc sheet, for trying Chapter 7

When you first save, RetroInv adds your inventory file, RETROINV.DAT unless you give it another name, and a small file called RETROINV.LAST that remembers which inventory you used last.

Make a Backup Copy First

Before you do anything else, make a copy of the RetroInv disk and put the original away. Use the System Utilities disk, Copy II Plus, or the Copy Disk command in the Apple II DeskTop. Then use the copy every day.



Warning: Your inventory lives on the same disk as the program. Copy the disk again from time to time, so that a worn floppy can't take your collection's records with it.

The Disk Starts by Itself

The RetroInv disk is a ProDOS startup disk. When you start your Apple with it, ProDOS starts RETROINV.SYSTEM automatically, because it's the first system program on the disk. BASIC.SYSTEM is there for the Applesoft version.

Starting RetroInv

There are three ways to start RetroInv:

- **From its own disk.** Put the disk in drive 1 and turn on your Apple, or press CONTROL-⌘-RESET.
- **From the Apple II DeskTop or another program selector.** Open the RetroInv disk or folder and double-click RETROINV.SYSTEM.
- **From Applesoft BASIC.** Type –/RETROINV/RETROINV.SYSTEM and press RETURN.



By the Way: You can copy RetroInv onto a hard disk. Put RETROINV.SYSTEM in its own folder. RetroInv keeps RETROINV.DAT and any Mouse Calc sheets in that same folder, even when you start it by its pathname while another folder is the prefix.

The Title Screen

First you'll see the 6502 Retro Computing title screen in color, with the words RETROINV 2.5 and COLLECTION INVENTORY. Press any key or click the mouse button to go on. If you do nothing, RetroInv goes on by itself after about five seconds.



Figure 1-1. The title screen

RetroInv then opens the inventory you used last time, or `RETROINV.DAT` the first time, and shows the main screen. The very first time, there's no inventory yet, so the list is empty and the status line reads `0 of 138 items`.

CHAPTER 2

A Guided Tour

The main screen shows your whole inventory as a list, one item per line. Everything else in RetroInv opens in a window on top of this list and closes back to it.

The Main Screen

From top to bottom, the main screen has five parts:

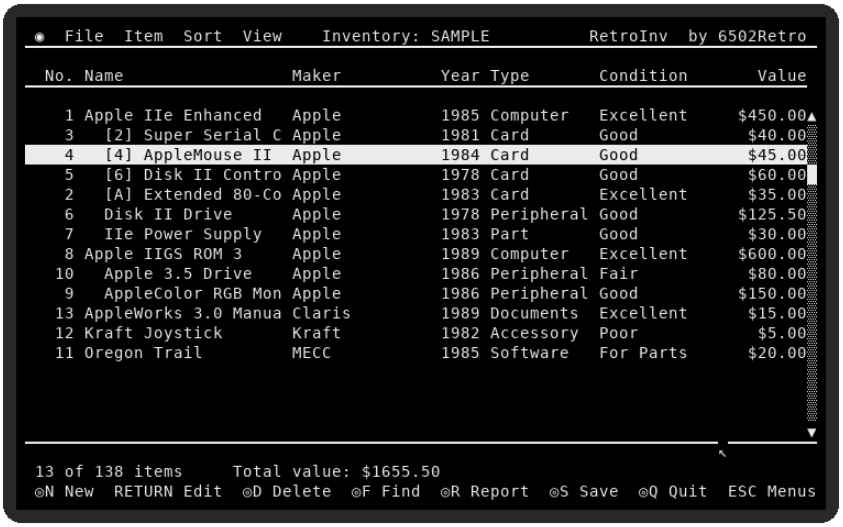


Figure 2-1. The main screen, showing the SAMPLE inventory sorted By System

1. **The menu bar**, along the top line. It holds five menus: the Apple menu (shown as a solid apple), File, Item, Sort and View. The name of the open inventory appears in the middle, and the words RetroInv by 6502Retro at the right end.
2. **The column headings**: No. (the Item No.), Name, Maker, Year, Type, Condition and Value. Cards and parts listed under their computer are indented, with the slot in brackets.
3. **The list**, with room for 17 items at a time. The selected item is shown in inverse (dark letters on a light bar).
4. **The status line**, which shows how many items you have out of the 138 that fit, such as 13 of 138 items, and the total value of the collection. When you've made changes that aren't saved yet, a diamond and the word Modified appear at the right.

5. **The help line** along the bottom, a reminder of the most-used commands.

Along the right edge of the list is the **scroll bar**. Its arrows and its gray track let you move through a long list with the mouse, and its light box shows where you are in the list.



By the Way: The serial number, location and notes don't fit on an 80-column line. To see them, open the item (see Chapter 3).

Using the Keyboard

PRESS	TO
↑ or ↓	Select the item above or below
← or →	Move up or down one screenful (17 items)
RETURN	Open the selected item to see or change it
ESC	Open the menus from the keyboard
⌘ and a letter	Give a command directly (see Appendix A)

Commands always need Open-Apple: press ⌘-N, not N alone. That way a stray key can't start a command, and Q on its own never quits.

Using the Mouse

If your Apple has a mouse, a pointer ↖ appears in the middle of the screen. Roll the mouse to move it.

- **Click** an item to select it. **Click it again** to open it.
- **Click** the up or down arrow of the scroll bar to move one item at a time. Click the gray track above or below the light box to move one screenful.
- **Click** a button in a window, such as OK or Cancel, to choose it.

Using the Menus

RetroInv's menus work like the ones in the Apple II DeskTop and Mouse Calc. There are three ways to choose a command:

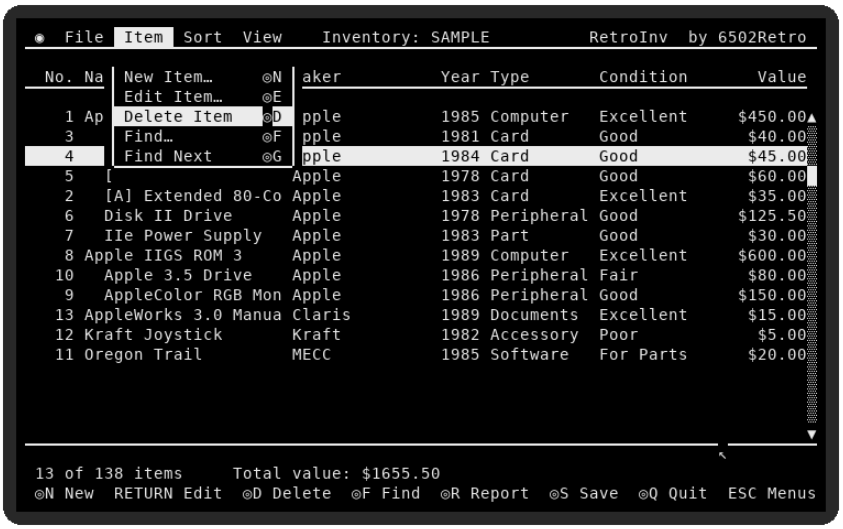


Figure 2-2. Choosing Delete Item from the Item menu

- **Drag.** Point at a menu title, press and hold the mouse button, move down to the command you want, and release the button.
- **Click.** Click a menu title once and the menu stays open. Then click the command you want. Clicking anywhere outside the menu closes it.
- **Keyboard.** Press ESC to open the menus. Use ← and → to move between menus and ↑ and ↓ to move within one, press RETURN to choose the highlighted command, or press ESC again to close the menus.

Commands with a keyboard shortcut show it at the right edge of the menu, such as ⌘S for Save.

MENU	COMMANDS
Apple	About RetroInv...
File	New Inventory, Open..., Save, Save As..., Revert to Saved, Import from Mouse Calc..., Export to Mouse Calc..., Quit
Item	New Item..., Edit Item..., Delete Item, Find..., Find Next
Sort	By Name, By Maker, By Year, By Type, By Value, By System
View	Report...

A command followed by three dots opens a window, where you can still change your mind before anything happens.

Windows and Buttons

RetroInv's windows have a title bar across the top and one or more buttons along the bottom. The button shown in inverse is the **default button**. Pressing RETURN chooses it, and ESC chooses Cancel. In a message window you can also press the first letter of a button's name, such as D for Don't Save.

In a window with several buttons, TAB or → moves the inverse highlight to the next button and ← moves it back, so RETURN chooses whichever button is highlighted.

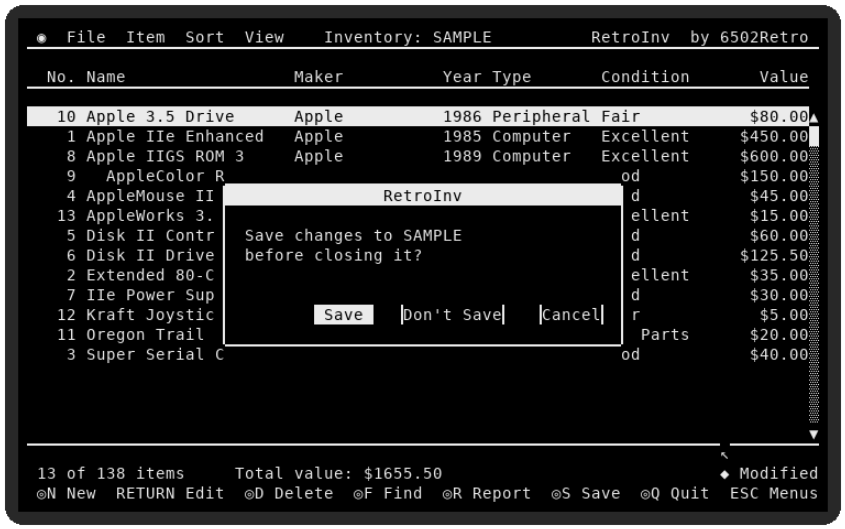


Figure 2-3. A message window. Save is the default button; TAB moves the highlight to Don't Save.

CHAPTER 3

Building Your Inventory

Each thing you own is one **item** in RetroInv: a computer, a disk drive, a game, a manual. This chapter shows you how to add items, change them and remove them.

Adding an Item

Choose New Item from the Item menu, or press ⌘-N. The New Item window opens with the cursor in the Name field.

1. Type the item's name, such as `Apple IIe Enhanced`, and press TAB.
2. Fill in the other fields the same way, pressing TAB after each one. You can leave any field except Name empty.
3. When you're done, press RETURN or click OK.

The new item appears at the bottom of the list, already selected, and the status line shows Modified.

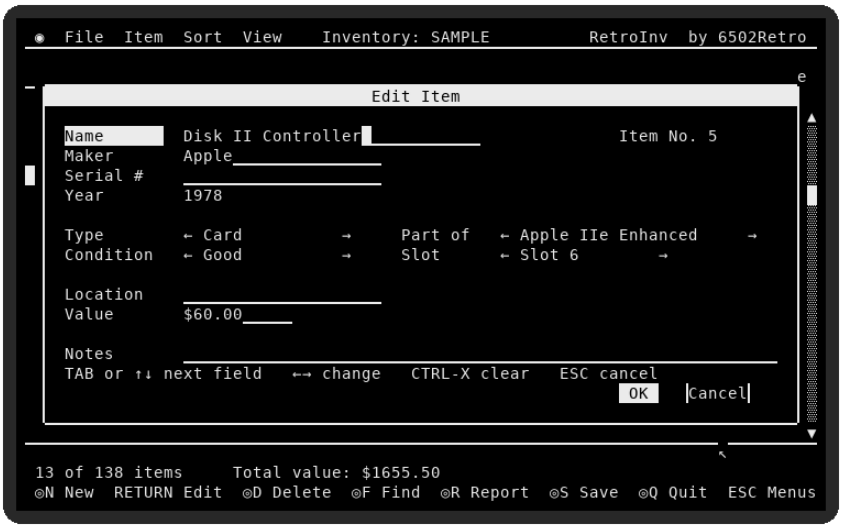


Figure 3-1. The item window for a Disk II controller in slot 6 of an Apple IIe



By the Way: You don't have to press TAB through every field. RETURN finishes the item from any field.

The Fields

FIELD	HOLDS	LONGEST ENTRY	WHAT TO TYPE
Name	What the item is	30 characters	Required. `Apple IIe Enhanced`, `Disk II`, `Karateka`
Maker	Who made it	20 characters	`Apple`, `Broderbund`, `Videx`
Serial #	The serial number	20 characters	Letters, digits and dashes are all fine
Year	When it was made	4 digits	Four digits from 1000 to 2999, such as `1983`
Type	What kind of item it is	One of eight	Computer, Card, Peripheral, Part, Software, Accessory, Documents, Other
Condition	What shape it's in	One of six	Mint, Excellent, Good, Fair, Poor, For Parts
Part of	The computer it belongs to	Any computer in the inventory	Not used for computers themselves
Slot	Where a card is plugged in	Slot 0 to 7, or the IIe's auxiliary slot	Used for cards in a computer
Location	Where you keep it	20 characters	`Shelf 2`, `Box A`, `Attic`
Value	What it's worth	Up to $9,999,999.99	Dollars and cents, such as `450` or `125.50`
Notes	Anything else	60 characters	`Platinum keyboard, 128K, works`

Every item also has an **Item No.**, shown at the top right of the window. RetroInv gives each new item the next free number, and it never changes, even when you sort the list or rename the item.

A new item starts out as a Computer in Good condition. To change Type, Condition, Part of or Slot, move to the field and press → or ← (or the Space bar) to step through the choices. With the mouse, click the arrow on either side.



Important! RetroInv doesn't accept the double quotation mark (") in any field, because it uses quotation marks inside its data file. Use an apostrophe (') instead.

Typing in a Field

PRESS	TO
TAB or ↓	Go to the next field
↑	Go back to the previous field
DELETE or ←	Erase the last character
CONTROL-X	Erase the whole field
RETURN	Finish (same as clicking OK)
ESC	Close the window without saving your changes (same as clicking Cancel)

With the mouse, click a field's name to move to it. The current field's name is shown in inverse, and a solid cursor shows where the next character will go.

Changing an Item

To change an item, select it and press RETURN, or click it twice, or choose Edit Item from the Item menu. The Edit Item window shows everything you've entered for the item, including the fields that don't fit on the main screen. Make your changes and press RETURN, or press ESC to leave the item as it was.

Deleting an Item

Select the item and choose Delete Item from the Item menu, or press ⌘-D. RetroInv shows the item's name and asks you to confirm. Click Delete or press RETURN to remove it, or click Cancel or press ESC to keep it.



Warning: There's no undo for Delete. If you delete an item by mistake and haven't saved since, choose Revert to Saved from the File menu to go back to your last saved inventory. That also throws away any other changes since you last saved.

Putting Items in a Computer

A computer and the cards, drives and parts that belong to it make up a **system**. To record that a card is in a computer:

1. Select the card and press RETURN to open it.
2. Move to Part of and press → until the computer's name appears. Only computers are offered.

3. Move to Slot and choose the slot the card is in: Slot 0 to Slot 7, or Aux slot for the auxiliary slot of an Apple IIe, where 80-column cards go.
4. Press RETURN.

Peripherals such as disk drives and monitors, and parts such as a power supply or a keyboard, can belong to a computer too. Leave their Slot set to None.

To see your systems, choose By System from the Sort menu (Chapter 4). Each computer is listed with its parts indented underneath, in slot order.



By the Way: To take an item out of a system, set its Part of back to None. Its Slot is kept, in case you put it back.

When you delete a computer, RetroInv tells you how many parts it has. They stay in the inventory, but no longer belong to any system. The same happens if you change a computer's Type to something else.



Important! A computer can't be part of another computer, so a computer's Part of field always shows None. A card that adds a computer to your Apple, such as a Z-80 card, is a Card.

How Many Items Fit

RetroInv holds your whole inventory in memory, up to 138 items, the same limit as the Applesoft version. The status line shows how many you have, for example `8 of 138 items`. When the inventory is full, New Item tells you so. Appendix C explains the limits.

CHAPTER 4

Finding and Arranging

When your collection grows past a screenful, two features help you find your way: Find, which jumps to the item you're looking for, and Sort, which puts the whole list in order.

Finding an Item

1. Choose Find from the Item menu, or press ⌘-F.
2. Type a word or part of a word, up to 20 characters. For example, type `MOUSE` to find an AppleMouse, or a serial number.
3. Press RETURN or click Find.

RetroInv looks through the Name, Maker, Serial #, Location and Notes of each item, starting just after the selected item. It selects the first item that contains what you typed and scrolls the list to show it. Capital and small letters count as the same, so `disk` finds Disk II.

To look for the next item that matches, choose Find Next from the Item menu, or press ⌘-G. When Find Next reaches the end of the list, it goes on from the top.

If no item matches, RetroInv tells you so and leaves the selection where it was.



By the Way: Find is a quick way to see everything in one place. Type `SHELF 2`, then press ⌘-G repeatedly to step through everything on that shelf.

Sorting the List

Choose one of the commands in the Sort menu to put the list in a new order:

COMMAND	ORDER
By Name	A to Z by name
By Maker	A to Z by maker
By Year	Oldest first

COMMAND	ORDER
By Type	Computers, then Cards, Peripherals, Parts, Software, Accessories, Documents and Other
By Value	Most valuable first
By System	Each computer, A to Z, followed by its parts; then the items that belong to no system

By System lists each computer's cards in slot order (slots 0 to 7, then the auxiliary slot), followed by its other parts. Parts are indented under their computer, and a card's slot appears in brackets: `[6]` for slot 6, `[A]` for the auxiliary slot.

Sorting ignores the difference between capital and small letters. After a sort, the first item in the new order is selected.

The new order is part of your inventory, so the status line shows Modified until you save. The next time you start RetroInv, the list is in the order you last saved.



By the Way: Items with no maker or year sort before those that have one.

A full inventory sorts in a few seconds on a 1 MHz Apple IIe. A message on the status line tells you RetroInv is busy.

The Collection Report

The Collection Report sums up your whole collection in one window: how many items of each type you have, what they're worth, and which items are the oldest and the most valuable.

To see it, choose Report from the View menu, or press ⌘-R.



Figure 5-1. The Collection Report

What the Report Shows

- **A table by type.** One line for each of the eight types, showing the number of items and their total value.
- **Total.** The number of items in the whole inventory and their combined value. This matches the total on the status line.
- **Oldest item.** The item with the earliest year, with the year in parentheses. Only four-digit years count, so an item with no year, or with a short year such as 85 from an old file, is left out.
- **Most valuable.** The item with the highest value, and that value.
- **In systems.** How many items belong to a computer, and how many computers there are.

When you've finished reading, press any key or click OK to go back to the list.



By the Way: The report is a good reason to fill in the Year and Value fields. Approximate values still make the totals useful, for insurance or for deciding what to show at the next swap meet.

If you'd like the report on paper, export your inventory to Mouse Calc (Chapter 7) and print it from there.

CHAPTER 6

Saving Your Work

RetroInv works on a copy of your inventory in the Apple's memory. Nothing you change is kept on disk until you save it, so save often.

Saving

Choose Save from the File menu, or press ⌘-S. RetroInv writes the whole inventory to its file in the RetroInv folder, and the Modified marker disappears from the status line. If the inventory doesn't have a name yet, RetroInv asks for one first, just as Save As does.

RetroInv never writes over your inventory directly. It saves into a new file called `RETROINV.SAVING` first, and only when that has worked does it replace the old file with it. If the save fails, for example because the disk is full, the file on disk stays exactly as it was at your last save. While it saves, the disk needs room for both copies.



Important! If an inventory file has more than 138 items, or doesn't hold the number of items it says it does, RetroInv tells you and opens it as Untitled. The file on disk is left alone. To keep what was loaded, choose Save As and give it a new name.
Important! Leave the RetroInv disk in the drive, unprotected, while you work. If the disk is write-protected or full, RetroInv tells you so and keeps your changes in memory, so you can fix the problem and save again.

Going Back to the Last Save

If you've made changes you don't want, choose Revert to Saved from the File menu. RetroInv asks you to confirm, then throws away every change since your last save and loads the inventory again. A new inventory that has never been saved has nothing to revert to, so the command does nothing.

Quitting

Choose Quit from the File menu, or press ⌘-Q. If you have unsaved changes, RetroInv asks whether to save them first:

- **Save** saves your inventory, then quits.
- **Don't Save** quits and throws away your changes.
- **Cancel** takes you back to RetroInv.

RetroInv then returns you to ProDOS, which starts the program you came from, such as the Apple II DeskTop.



Warning: Turning off the Apple or pressing CONTROL-⌘-RESET while RetroInv is running loses any changes you haven't saved.

Keeping More Than One Inventory

You can keep as many inventories as you like, such as one for Apple equipment and one for Commodore, or one for the collection and one for items for sale. Each inventory is its own file in the RetroInv folder. The menu bar always shows which one is open: Inventory: COMMODORE, for example.

To save an inventory under a new name, choose Save As from the File menu. Type a name and press RETURN or click Save. From then on, the inventory with the new name is the one that's open; the file with the old name stays on the disk as it was at your last save.

To start a new inventory, choose New Inventory from the File menu. The list empties and the menu bar shows Inventory: Untitled. The first time you save, RetroInv asks for a name.

To switch to another inventory, choose Open from the File menu, or press ⌘-O. RetroInv lists the inventories in its folder. Select one with ↑ and ↓ and press RETURN, or click it twice.

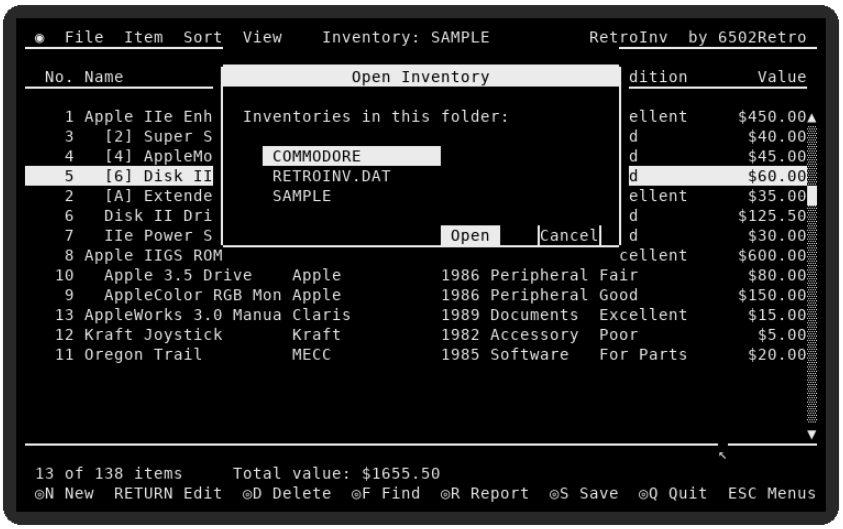


Figure 6-1. Choosing an inventory to open

If you have unsaved changes when you choose New Inventory, Open or Quit, RetroInv first asks whether to save them: Save, Don't Save or Cancel.

RetroInv remembers the inventory you were using and opens it again the next time you start the program. If that file has been deleted, or isn't an inventory, it opens `RETROINV.DAT` instead.



By the Way: An inventory's name follows the ProDOS rules: it starts with a letter and has up to 15 letters, digits and periods. RetroInv won't let you use the name of a file that isn't an inventory, such as a Mouse Calc sheet, and it asks before replacing another inventory. **Important!** Open lists the inventories in the RetroInv folder only, up to 14 of them. Other text files in the folder don't take up places. If there are more than 14 inventories, the window says Only the first 14 are listed. Keep your inventories in the same folder as `RETROINV.SYSTEM`.

The Applesoft Version

The RetroInv disk also holds `RETROINV`, a version of the program written in Applesoft BASIC. It doesn't have menus, a mouse pointer or Mouse Calc import and export, but it runs on every Apple II with 48K of memory, including the original Apple II and the II Plus, in 40 columns.

Both versions can read and write `RETROINV.DAT`, so you can enter items at one machine and look them up on another. The Applesoft version always uses `RETROINV.DAT`; it can't open inventories saved under other names.

To use it under ProDOS, start `BASIC.SYSTEM` and type:

```
PREFIX /RETROINV
RUN RETROINV
```

The Applesoft version is also available on a DOS 3.3 disk. Under DOS 3.3, type `RUN RETROINV`.

The Applesoft version shows a numbered menu: Add an Item, Browse / Edit / Delete, List All Items, Search, Sort by Name, Collection Report, Save to Disk and Quit. Type the number of the one you want. It's written to work with Applesoft compilers such as TASC and the Beagle Compiler, if you want it to run faster.

The Applesoft version knows about systems too. When you add or edit an item that isn't a computer, it asks which computer the item is part of: type that computer's Item No., shown when you browse it, or 0 for none. Then press a slot number from 0 to 7, A for the auxiliary slot, or N for none. Browsing an item shows its Item No., the computer it belongs to and its slot.

Browsing shows each item's Item No. at the top, with its place in the list after it, such as `ITEM NO. 17 (3 OF 20)`. J)ump in Browse and the View question after a Search both ask for an Item No. Search treats capital and small letters as the same.

Applesoft RetroInv 1.4 holds up to 138 items, the same as RetroInv 2.5, so a file saved by either version opens completely in the other. Applesoft keeps text in its own memory, so a collection with long names and notes can fill memory first. Then the Applesoft version says `NOT ENOUGH MEMORY FOR MORE ITEMS` instead of adding one.

If a file has more items than the Applesoft version can load, or is damaged, it loads what it can and asks for a new file name when you save, so the original file is never shortened. Like RetroInv, it saves into `RETROINV.SAVING` first and replaces the old file only when that has worked.



Important! Use the versions that came together: RetroInv 2.5 with Applesoft RetroInv 1.4. Older versions can't read the serial numbers, systems and slots that newer ones save.

CHAPTER 7

RetroInv and Mouse Calc

RetroInv can save your inventory as a Mouse Calc spreadsheet, and read one back. In Mouse Calc you can print your inventory, work out what-if totals, or enter a stack of new items quickly, then bring the result back into RetroInv.

Exporting Your Inventory

1. Choose Export to Mouse Calc from the File menu, or press ⌘-X.
2. Type a name for the sheet, or keep the suggested name, `INVENTORY`.
3. Press RETURN or click Export.

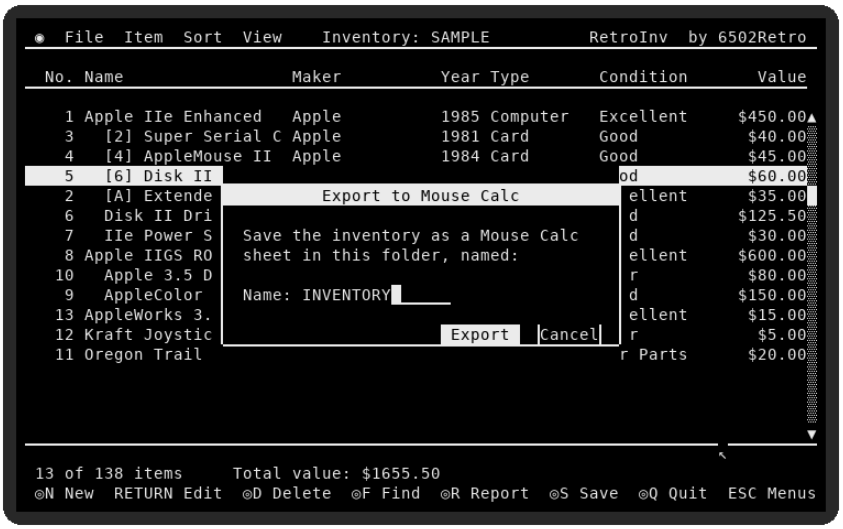


Figure 7-1. Naming the exported sheet

RetroInv saves the sheet in the RetroInv folder and tells you its full pathname, such as `/RETROINV/INVENTORY`. Exporting doesn't change your inventory or `RETROINV.DAT`.

If a Mouse Calc sheet with that name already exists, RetroInv asks before replacing it.



By the Way: A sheet's name follows the ProDOS rules: it starts with a letter and has up to 15 letters, digits and periods. RetroInv turns small letters into capitals as you type.

What the Sheet Looks Like

The exported sheet has one row for each item and one column for each field, with the column widths already set. Row 1 holds the headings:

COLUMN	HEADING	EXAMPLE	NOTES
A	Name	Disk II Controller	
B	Maker	Apple	
C	Serial #		
D	Year	1978	A number when it's all digits
E	Type	Card	
F	Condition	Good	
G	Location		
H	Value	$60.00	Mouse Calc's $X,XXX.XX format
I	Notes		
J	Slot	Slot 6	Or Aux slot; blank when there's none
K	Part Of	1	The Item No. of the computer it belongs to
L	Item No.	5	

Below the last item, a Total row holds `+SUM(H2..Hn)` under Value.

Values use Mouse Calc's $X,XXX.XX format, and years are numbers. The Total row holds a real Mouse Calc formula, so it keeps adding up correctly when you change values in Mouse Calc.

Working in Mouse Calc

Start Mouse Calc, choose Open, and open the sheet from the RetroInv folder. From there it's an ordinary Mouse Calc sheet: you can print it, sort it, add columns of your own, or type in new items in the empty rows.

If you plan to import the sheet back into RetroInv, follow three rules:

- **Keep the headings in row 1.** RetroInv finds each column by its heading. Capitals, spaces and punctuation don't matter, so Serial #, SERIAL and serial all work, but the heading must be the whole name: a column headed Name of item, Part No. or Year bought is ignored. The columns can be in any order.

- **Give every item a name.** Rows with an empty Name column are skipped, which is how RetroInv leaves out blank rows and the Total row.
- **Save the sheet in the RetroInv folder,** so RetroInv can find it.

Columns RetroInv doesn't recognize, such as one you add for Bought From, are simply ignored on import.

Importing a Sheet

1. Choose Import from Mouse Calc from the File menu, or press ⌘-I.
2. RetroInv lists the Mouse Calc sheets in its folder. Select one with ↑ and ↓ and press RETURN, or click it twice.
3. RetroInv asks you to confirm. Click Import or press RETURN.

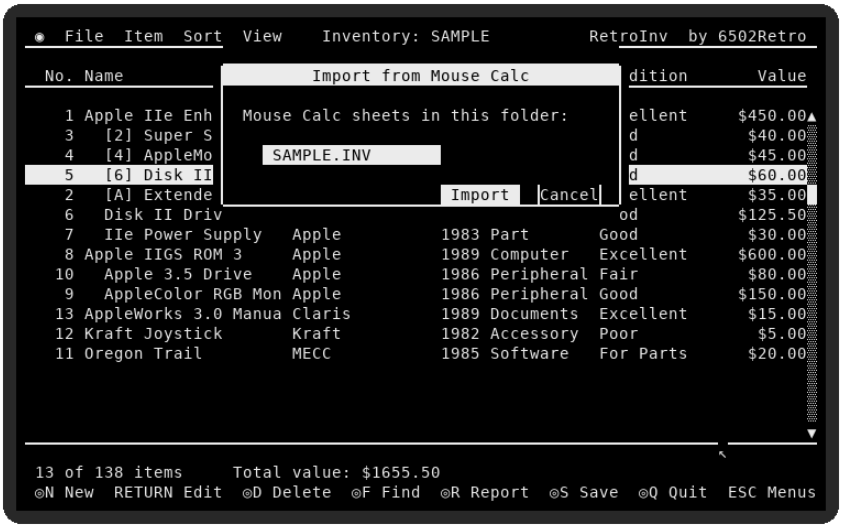


Figure 7-2. Choosing a sheet to import



Warning: Importing replaces the whole inventory in memory with the sheet; it doesn't add the sheet to your items. Your disk isn't changed until you save, so if you import the wrong sheet, choose Revert to Saved to bring back your saved inventory. To keep the imported items separately, choose Save As and give them their own name.

RetroInv tells you how many items it read. Look the list over, then choose Save to keep them.

How RetroInv Reads a Sheet

COLUMN	RETROINV ACCEPTS
Name, Maker, Serial #, Location, Notes	Text or numbers. Long entries are cut to fit the field. Double quotation marks become apostrophes.
Year	A number or text, up to 4 characters
Type	A name (Computer, Card, Peripheral, Part, Software, Accessory, Documents, Other) or a number from 1 to 8. The first three letters of a name are enough. A number is the file code from Appendix C (2 is Peripheral, 7 is Card), not the item's place in the Type list.
Condition	A name (Mint, Excellent, Good, Fair, Poor, For Parts) or a number from 1 to 6
Value	A number, or text such as $1,200.00. Negative values are read as zero, and anything over $9,999,999.99 as $9,999,999.99.
Slot	Slot 0 to 7 as a number or text such as Slot 6, or any word starting with A for the auxiliary slot
Part Of	The Item No. of a computer in the same sheet
Item No.	A number. Rows without one, with a number used twice, or with a number over 65,535 get a new one.

Part Of works only when the sheet also has an Item No. column, so it can find the computer. A Part Of that doesn't match any computer in the sheet is dropped.

A Type or Condition RetroInv doesn't recognize becomes Other or Good. If the sheet has more rows than fit in memory, RetroInv reads as many as it can and tells you.



By the Way: Try it with `SAMPLE.INV` on the RetroInv disk. Import it, look at the list, then choose Revert to Saved to put your own inventory back.

APPENDIX A

Command Summary

Every RetroInv command, with its menu and keys. On the main screen, commands always need Open-Apple.

Menu Commands

COMMAND	MENU	KEYS	WHAT IT DOES
About RetroInv...	Apple	⌘-?	Shows the version and author
New Inventory	File		Starts an empty, untitled inventory
Open...	File	⌘-O	Switches to another inventory in the RetroInv folder
Save	File	⌘-S	Saves the open inventory, asking for a name if it has none
Save As...	File		Saves the inventory under a new name
Revert to Saved	File		Throws away changes since the last save
Import from Mouse Calc...	File	⌘-I	Replaces the inventory with a Mouse Calc sheet
Export to Mouse Calc...	File	⌘-X	Saves the inventory as a Mouse Calc sheet
Quit	File	⌘-Q	Leaves RetroInv, offering to save first
New Item...	Item	⌘-N	Adds an item
Edit Item...	Item	⌘-E or RETURN	Shows and changes the selected item
Delete Item	Item	⌘-D	Removes the selected item, after you confirm
Find...	Item	⌘-F	Finds an item by any part of its text

COMMAND	MENU	KEYS	WHAT IT DOES
Find Next	Item	⌘-G	Finds the next item that matches
By Name, By Maker, By Year, By Type, By Value, By System	Sort		Puts the list in that order
Report...	View	⌘-R	Shows totals for the collection

Keys on the Main Screen

KEY	ACTION
↑ ↓	Select the previous or next item
← →	Move up or down one screenful
RETURN	Edit the selected item
ESC	Open the menus

Keys in the Menus

KEY	ACTION
← →	Move to the menu on the left or right
↑ ↓	Highlight a command
RETURN	Choose the highlighted command
ESC	Close the menus

Keys in the Item Window

KEY	ACTION
TAB or ↓	Next field
↑	Previous field
← → or Space	Change Type, Condition, Part of or Slot
DELETE or ←	Erase a character (in a text field)
CONTROL-X	Erase the field
RETURN	OK
ESC	Cancel

Keys in Message Windows

KEY	ACTION
RETURN	The highlighted (inverse) button
TAB or →	Highlight the next button
←	Highlight the previous button
ESC	Cancel, or the last button
First letter of a button	That button

The Mouse

TO	DO THIS
Choose a menu command	Drag down the menu and release, or click the title, then the command
Select an item	Click it
Edit an item	Click it again once it's selected
Scroll one line	Click a scroll arrow
Scroll one screenful	Click the gray track
Move to a field	Click the field's name
Change Type or Condition	Click its ← or → side

Messages

These are all the messages RetroInv can show, in alphabetical order, with what each one means and what to do.

MESSAGE	WHAT IT MEANS	WHAT TO DO
A Mouse Calc sheet with that name already exists. Replace it?	You're exporting over an existing sheet	Click Replace to overwrite it, or Cancel and choose another name
A ProDOS name starts with a letter and has up to 15 letters, digits and periods.	The name you typed for an inventory or sheet isn't allowed	Type a name that starts with a letter
An inventory with that name already exists. Replace it?	Save As would write over another inventory	Click Replace to overwrite it, or Cancel and choose another name
Could not export the sheet.	The disk couldn't be written	See the line that follows the message, and the disk errors below
Could not open the inventory.	The inventory file couldn't be read	See the disk errors below. If you were opening another inventory, the one you had open stays open
Could not save the inventory. ... The file on disk was not changed.	The inventory couldn't be written	See the disk errors below. Your changes are still in memory and the saved file is as it was at your last save. A failed Save As keeps the old name
Delete this item?	You chose Delete Item	Click Delete, or Cancel to keep the item. For a computer, it also says how many of its parts will stay in the inventory, unattached
Discard your changes and reload the inventory from disk?	You chose Revert to Saved	Click Revert to go back to your last save, or Cancel
Disk I/O error.	The disk couldn't be read or written	Check that the disk is in the drive and the door is closed. Try a fresh copy of the disk

MESSAGE	WHAT IT MEANS	WHAT TO DO
Exported N items to...	The export worked	Open the sheet in Mouse Calc
Imported N items. Choose Save to keep them.	The import worked	Look over the list, then save
NEEDS ENHANCED IIE/IIC/IIGS - PRESS KEY	Your Apple can't show MouseText	Use the Applesoft version (Chapter 6)
No item matches that text.	Find found nothing	Try fewer letters, or check the spelling
Only the first 14 are listed.	Open or Import found more than 14 files to list	Move some inventories or sheets to another folder
Please give the item a name.	You clicked OK with the Name field empty	Type a name, or press ESC to cancel
Please type the year with four digits, like 1984, or leave it blank.	The Year isn't four digits from 1000 to 2999	Type the whole year, or erase the field with CONTROL-X
ProDOS error $nn	A disk error not listed here	Note the number. Check the disk and drive, and try again
Row 1 of that sheet has no Name column...	The sheet doesn't look like an inventory	Add a Name heading in row 1 in Mouse Calc, or choose another sheet
Save changes to NAME before closing it?	You chose New Inventory, Open or Quit with unsaved changes	Click Save, Don't Save, or Cancel. TAB moves the highlight between them
Some rows did not fit in memory.	An imported sheet had more than 138 rows	The first rows were read. Split the sheet into two to keep everything
That name belongs to a file that is not a Mouse Calc sheet.	You're exporting over a file of another kind, or using one of RetroInv's own names	Choose another name
That name belongs to a file that is not an inventory.	Save As would write over a file of another kind, or over RETROINV.LAST	Choose another name
The disk is full.	There's no room left on the disk	Delete files you don't need, or use a disk with more room. A save needs room for a second copy of the inventory while it works
The disk is write-protected.	The disk can't be written	On a 5.25-inch disk, uncover the write-protect notch. On a 3.5-inch disk, slide the tab to cover the hole

MESSAGE	WHAT IT MEANS	WHAT TO DO
The inventory is full.	The inventory already has 138 items	Start a second inventory with New Inventory
The largest value RetroInv can keep is $9,999,999.99.	The Value you typed is too large	Type a smaller value
There are no inventories in this folder yet.	Open found nothing to list	Choose Save As to start one
There are no Mouse Calc sheets in this folder.	Import found nothing to list	Copy a sheet into the RetroInv folder, or export one first
This file doesn't hold the number of items it says...	The file ends early, or has more items than its count, so it may be damaged	It's open as Untitled and the file is left alone. Look over the items, then use Save As to save them under a new name
This inventory has more than 138 items...	The file holds more items than RetroInv can	The first 138 are open as Untitled and the file is left alone. Use Save As to keep them under a new name

APPENDIX C

Technical Information

This appendix is for the curious and for programmers. You don't need it to use RetroInv.

Limits

LIMIT	RETROINV 2.5	APPLESOFT RETROINV 1.4
Items per inventory	138	138, or fewer if long text fills memory
Inventories	As many as the disk holds; Open lists 14	RETROINV.DAT, or a new name when a file wasn't loaded in full
Item No.	1 to 65,535	1 to 65,535
Name	30 characters	30 characters
Maker, Serial #, Location	20 characters each	20 characters each
Notes	60 characters	60 characters
Year	Blank, or 4 digits from 1000 to 2999	Blank, or 4 digits from 1000 to 2999
Value	`$0.00` to `$9,999,999.99`, kept to the cent	`$0.00` to `$9,999,999.99`, rounded to the cent
Total value	Exact for 138 items at the largest value	Exact for 138 items at the largest value
Find text	20 characters	20 characters
Mouse Calc sheets listed by Import	14	none

Memory Map

RetroInv is a ProDOS 8 system program. ProDOS loads it at `$2000`, where its first 8K is the hi-res title picture, displayed straight from where it was loaded. After the title screen, that memory is reused for items.

ADDRESS	USE
$0300-$03CF	ProDOS prefix, Mouse Calc pathname and column map
$0400-$07FF	80-column text screen, main and auxiliary memory
$0800-$0913	Item pointer tables
$0914-$09F3	Names for the Open and Import lists
$0A00-$0BFF	File and directory buffers
$0C00-$3F6F	Item storage (77 items), in the title picture's memory
$3F70-$3FFF	Work buffers
$4000-$91D4	Program code, text and work areas
$91D5-$BAFF	More item storage (61 items)
$BB00-$BEFF	ProDOS file buffer

Each item takes 171 bytes. Strings are stored with a length byte, and the value is a 32-bit count of cents. Totals are added up in 40 bits, so even 138 items at the largest value total exactly $1,379,999,998.62.

Inventory Files

Each inventory, `RETROINV.DAT` or any name you give with Save As, is a ProDOS text file (type `$04`), readable by any program that reads text. Its first line is `RETROINV3`, which is how Open recognizes it; the second is the number of items, and then each item takes twelve lines:

```
"Disk II Controller"
"Apple"
"1978"
7
3
""
60.00
""
""
5
1
7
```

In order, the lines are Name, Maker, Year, Type, Condition (1-6), Location, Value, Notes, Serial #, Item No., Part Of and Slot. Type is 1 Computer, 2 Peripheral, 3 Software, 4 Accessory, 5 Documents, 6 Other, 7 Card or 8 Part. Part Of is the Item No. of the

computer, or 0 for none. Slot is 0 for none, 1 to 8 for slots 0 to 7, or 9 for the auxiliary slot, so the example above is a Disk II controller in slot 6 of item 1. Text is in quotation marks so that Applesoft's INPUT statement can read commas and colons.

Both versions write Value in dollars and cents, such as `1200.00`, never in Applesoft's E notation. When they read a file, they also accept E notation such as `1E-03` or `1.5E+03`, round it to the cent, and treat anything over $9,999,999.99 as $9,999,999.99. A Year that isn't four digits is kept as it is, but it doesn't count as the oldest year in the report.

Both current versions also read the older formats: `RETROINV2` files, with nine lines per item and no Item No., Part Of or Slot, and the original files with no first line, eight lines per item and no Serial #. Items from older files are numbered in order, and the file is written in the new format the next time you save.

`RETROINV.LAST` is a one-line text file holding the name of the inventory to open at startup. RetroInv rewrites it after Open and Save As. If the disk is write-protected, RetroInv simply opens `RETROINV.DAT` next time.

If `RETROINV.LAST` names a file that isn't an inventory, such as a Mouse Calc sheet, RetroInv ignores it and opens `RETROINV.DAT`.

`RETROINV.SAVING` exists only while a save is in progress. If you ever find one, a save was interrupted, for example by turning the Apple off. It is a complete inventory if the save got that far: open it with Open to look, then Save As to give it a proper name.

Mouse Calc Sheets

Mouse Calc sheets are ProDOS files of type `$F8`, auxiliary type `$0000`. Each one starts with a block of 2,179 bytes that records Mouse Calc's window settings and column widths, followed by the cells, column by column. Numbers are stored in Mouse Calc's own decimal floating-point format, which holds 12 significant digits.

RetroInv's exported sheets use the settings block of a sheet saved by Mouse Calc 1.1, with RetroInv's column widths filled in. They're the same, byte for byte, as the sheets made by the Mouse Calc Converter.

The Mouse

RetroInv looks for mouse firmware in slots 7 down to 1 when it starts, and uses the first mouse it finds. It reads the mouse directly and doesn't use interrupts, so it works alongside any other card. If there's no mouse, RetroInv runs from the keyboard alone.

Compiling the Applesoft Version

The Applesoft version doesn't jump into the middle of loops or subroutines, and it dimensions every array with a constant size, so TASC and the Beagle Compiler can compile it unchanged. Its disk error handling uses `ONERR GOTO`, `PEEK(222)` and `POKE 216,0`.

APPENDIX D

Troubleshooting

If something doesn't work the way this manual says, look for your problem here.

PROBLEM	TRY THIS
The screen shows strange letters and symbols instead of windows	Your Apple isn't enhanced, or the 80-column card is missing. Use the Applesoft version
There's no mouse pointer	RetroInv didn't find a mouse when it started. Check that the AppleMouse card is seated and the mouse is plugged in, then start RetroInv again. Meanwhile, use the keyboard
The pointer jumps or sticks	Clean the mouse ball and the rollers inside the mouse, and use it on a clean, flat surface
Pressing a letter does nothing	Commands need Open-Apple. Hold ⌘ and press the letter
My items are gone when I start RetroInv	RetroInv opens the inventory you used last, which may not be the one you expect. Check the name on the menu bar, choose Open from the File menu to pick another, and make sure you saved before quitting
The menu bar says Untitled right after opening a file	The file had more than 138 items or looked damaged, so RetroInv left it alone. See Appendix B, then use Save As
Save fails every time	The disk is probably write-protected or full. A save needs room for a second copy while it works. See Appendix B
A file called RETROINV.SAVING is on the disk	A save was interrupted. See Inventory Files in Appendix C
Import says there are no Mouse Calc sheets	The sheet isn't in the RetroInv folder. In Mouse Calc, save it into that folder
Imported items are missing	Rows without a Name are skipped. Check that every item in the sheet has a name and that row 1 holds the headings
An imported column is empty	Its heading in row 1 isn't one RetroInv knows. Rename it to the heading in Chapter 7, such as Notes or Part Of
Mouse Calc doesn't show the exported sheet	In Mouse Calc's Open window, open the RetroInv folder. Sheets are saved next to RETROINV.SYSTEM

PROBLEM	TRY THIS
Old files lose their serial numbers	Use RetroInv 2.5 or Applesoft RetroInv 1.4 on every machine

Glossary

TERM	MEANING
80-column card	A card that lets the Apple IIe show 80 characters across the screen. The IIc and IIGS have one built in
Auxiliary slot	The extra slot inside an Apple IIe that holds an 80-column card
Click	To press and quickly release the mouse button
Default button	The button shown in inverse in a window. Pressing RETURN chooses it
Drag	To hold down the mouse button while moving the mouse
Enhanced Apple IIe	An Apple IIe with the 1985 enhancement, which adds MouseText. The startup screen says Apple //e
Export	To save your inventory in a form another program can use
Field	One piece of information about an item, such as its Name or Year
Folder	A ProDOS directory that holds files. Also called a subdirectory
Import	To read information that another program saved
Inverse	Dark characters on a light background, used to show what's selected
Item	One thing in your collection
Item No.	The permanent number RetroInv gives each item. Part Of refers to a computer by its Item No.
Menu bar	The line across the top of the screen that holds the menu titles
Mouse Calc	A spreadsheet program for the Apple II from Version Soft
MouseText	Special characters, such as the pointer and window borders, built into the enhanced IIe, IIc and IIGS
Open-Apple key	The key with the outlined apple, used with letters for commands
Pathname	The full name of a file, including its disk and folders, such as /RETROINV/INVENTORY
ProDOS	The Apple II disk operating system that RetroInv uses
Prefix	The folder ProDOS uses when a name doesn't say which folder

TERM	MEANING
Scroll bar	The bar at the right of the list, used to move through it with the mouse
Sheet	A Mouse Calc spreadsheet file
Slot	One of the expansion connectors inside an Apple II, numbered 0 to 7, where cards plug in
Status line	The line near the bottom of the screen that shows the item count and total value
System	A computer together with the cards, peripherals and parts that are part of it
Write-protect	To protect a disk so nothing can be written on it



RetroInv 2.5 for ProDOS · Applesoft RetroInv 1.4

6502RETRO · RETRO COMPUTING